Chapter 1. Probability Distributions

Axioms, conditional probability, random variables, moments, generating functions, inequalities

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1.1 Random experiments and probability

A **random experiment** has outcomes that cannot be predicted in advance; its **sample space** S is the set of all outcomes, and **events** are subsets of S. When all N outcomes of a finite S are equally likely, $P(A) = \#A/N$ (classical probability); in general a probability is any assignment obeying Kolmogorov's axioms:

(A1) $P(A) \geq 0$ for every event A;

(A2) $P(S) = 1$;

(A3) (countable additivity) if $A_1, A_2, \ldots$ are pairwise disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem 1.1 (Basic properties)

(a) $P(\emptyset) = 0$, and finite additivity holds.

(b) $P(A^c) = 1 - P(A)$, and $0 \leq P(A) \leq 1$.

(c) (Monotonicity) $A \subseteq B \Rightarrow P(A) \leq P(B)$, and $P(B \setminus A) = P(B) - P(A)$.

(d) (Addition rule) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; more generally, inclusion–exclusion

$$P\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n).$$

(e) (Countable subadditivity) $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (a) Apply (A3) to $S \cup \emptyset \cup \emptyset \cup \cdots$: $1 = 1 + \sum P(\emptyset)$ forces $P(\emptyset) = 0$; padding a finite union with empty sets gives finite additivity.

(b) $S = A \cup A^c$ disjointly.

(c) $B = A \cup (B \setminus A)$ disjointly.

(d) $A \cup B = A \cup (B \setminus (A \cap B))$ disjointly and $P(B \setminus (A \cap B)) = P(B) - P(A \cap B)$ by (c); the general formula follows by induction on n, applying the two-set rule to $(A_1 \cup \cdots \cup A_{n-1}) \cup A_n$.

(e) Put $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$; these are disjoint with the same union and $B_n \subseteq A_n$. ■

Theorem 1.2 (Continuity of probability)

If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$; if $B_1 \supseteq B_2 \supseteq \cdots$ then $P(\bigcap_n B_n) = \lim_n P(B_n)$.

Proof. With $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1}$ (disjoint, same union),

$$P\Big(\bigcup_n A_n\Big) = \sum_k P(C_k) = \lim_n \sum_{k \leq n} P(C_k) = \lim_n P(A_n).$$

For decreasing B_n, apply this to $A_n = B_n^c$ and take complements. ■

Example 1.1

Let $P((a, b]) = b - a$ for subintervals of $[0, 1]$ (the uniform probability). For a point b, $\{b\} = \bigcap_n (b - \frac{1}{n}, b]$, so $P(\{b\}) = \lim_n \frac{1}{n} = 0$: every single point has probability zero, although some point always occurs. Likewise $P([a, b]) = P((a, b)) = b - a$.

Example 1.2

Roll three fair dice. $P(\text{at least one 6}) = 1 - P(\text{no 6}) = 1 - (5/6)^3 = 91/216 \approx 0.42$. Inclusion–exclusion gives the same: $3 \cdot \frac{1}{6} - 3 \cdot \frac{1}{36} + \frac{1}{216} = \frac{91}{216}$.

1.2 Conditional probability and independence

Definition (Conditional probability)

For $P(A) > 0$, $P(B \mid A) = P(A \cap B)/P(A)$. As a function of B it satisfies (A1)–(A3), so it is a probability on the reduced sample space A.

Theorem 1.3

(a) (Multiplication) $P(A \cap B) = P(B \mid A)P(A)$, and $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$.

(b) (Total probability) If $A_1, A_2, \ldots$ partition S with $P(A_i) > 0$, then $P(B) = \sum_i P(B \mid A_i)P(A_i)$.

(c) (Bayes) In the same setting, with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)}.$$

Proof. (a) is the definition, iterated.

(b) $B = \bigcup_i (B \cap A_i)$ disjointly; apply (A3) and (a).

(c) $P(A_j \mid B) = P(B \cap A_j)/P(B)$; expand the numerator by (a) and the denominator by (b). ∎

$P(A_j)$ are **prior** and $P(A_j \mid B)$ **posterior** probabilities.

Example 1.3 (Diagnostic test)

A disease has prevalence 1%; a test detects it with probability 0.95 and gives false positives with probability 0.05. Then

$$P(\text{disease} \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0095}{0.059} \approx 0.16.$$

A positive result raises the probability only from 1% to 16% because healthy people vastly outnumber sick ones.

Example 1.4 (Three machines)

Machines A, B, C make $50\%, 30\%, 20\%$ of output with defect rates $2\%, 3\%, 5\%$. Then $P(\text{defective}) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.029$ and $P(\text{C} \mid \text{defective}) = 0.010/0.029 \approx 0.34$.

Definition (Independence)

A, B are independent if $P(A \cap B) = P(A)P(B)$ (equivalently $P(B \mid A) = P(B)$ when $P(A) > 0$). $A_1, \ldots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every subcollection.

If A, B are independent, so are A, B^c (and A^c, B^c): $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)P(B^c)$. Pairwise independence does not imply mutual independence:

Example 1.5

Toss two fair coins: $A = \{\text{first H}\}$, $B = \{\text{second H}\}$, $C = \{\text{the two agree}\}$. Each has probability $\frac{1}{2}$ and each pair intersects with probability $\frac{1}{4}$, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

Example 1.6 (Parallel system)

A device works if at least one of three independent components works; each fails with probability 0.1. $P(\text{works}) = 1 - 0.1^3 = 0.999$.

1.3 Random variables and their distributions

A **random variable** X is a real-valued function on S; events such as $\{X \le x\} = \{s : X(s) \le x\}$ inherit probabilities, and the probabilities $P(X \in A)$ form the **distribution** of X.

- **Discrete** X takes values $x_1, x_2, \ldots$; its probability mass function (pmf) $f(x) = P(X = x)$ satisfies $f \ge 0$, $f = 0$ off $\{x_k\}$, $\sum_x f(x) = 1$, $P(a \le X \le b) = \sum_{a \le x \le b} f(x)$.
- **Continuous** X has a probability density function (pdf) $f \ge 0$ with $\int f = 1$ and $P(a \le X \le b) = \int_a^b f(x)\,dx$. Then $P(X = a) = 0$ and $P(a \le X \le a + \Delta) \approx f(a)\Delta$, so $f(a)$ is a probability per unit length, not a probability.

We write $X \sim f$; the indicator $I_A(x)$ gives compact formulas such as $f(x) = 2x I_{[0,1]}(x)$.

Example 1.7

Toss a fair coin until the first head; the number of tosses X has $f(k) = (1/2)^k$, $k = 1, 2, \ldots$, and $\sum_k f(k) = 1$. Two tosses give Y = number of heads with $f_Y(0) = \frac{1}{4}$, $f_Y(1) = \frac{1}{2}$, $f_Y(2) = \frac{1}{4}$.

Example 1.8

If $f(x) = cx(2 - x)$ on $[0, 2]$, then $\int_0^2 x(2 - x)\,dx = \frac{4}{3}$, so $c = \frac{3}{4}$ and $P(X \le \frac{1}{2}) = \frac{3}{4}\int_0^{1/2}(2x - x^2)\,dx = \frac{3}{4}(\frac{1}{4} - \frac{1}{24}) = \frac{5}{32}$.

1.4 Expectation, variance and moments

Definition (Expectation)

$E[g(X)] = \sum_x g(x)f(x)$ or $\int g(x)f(x)\,dx$, defined when the sum or integral converges absolutely. $\mu = E(X)$ is the mean (the distribution's center of mass), $\mathrm{Var}(X) = E[(X - \mu)^2]$ the variance and $\mathrm{sd}(X) = \sqrt{\mathrm{Var}\,X}$ the standard deviation.

Theorem 1.4 (Properties of expectation)

(a) $E(aX + b) = aE(X) + b$.
(b) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E g_1(X) + c_2 E g_2(X)$.

(c) $g_1 \le g_2 \Rightarrow Eg_1(X) \le Eg_2(X)$.

Proof. Linearity and monotonicity of sums and integrals. ■

Theorem 1.5 (Variance)

(a) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$.
(b) $\operatorname{Var} X = E(X^2) - (EX)^2$.

Proof. (a) $aX + b - E(aX+b) = a(X-\mu)$, so $\operatorname{Var}(aX+b) = E[a^2(X-\mu)^2]$.

(b) $E(X-\mu)^2 = E(X^2) - 2\mu EX + \mu^2 = E(X^2) - \mu^2$. ■

The **standardized** variable $Z = (X-\mu)/\sigma$ has $EZ = 0$, $\operatorname{Var} Z = 1$. A mean need not exist: for $f(k) = \frac{1}{k(k+1)}$, $k \ge 1$, $\sum k f(k) = \sum \frac{1}{k+1} = \infty$.

Example 1.9

For the uniform density on $[0,1]$, $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. For the triangular density $f(x) = 4x$ on $[0, \frac{1}{2}]$ and $4 - 4x$ on $[\frac{1}{2}, 1]$ (same mean), $EX^2 = \frac{7}{24}$ and $\sigma^2 = \frac{7}{24} - \frac{1}{4} = \frac{1}{24}$: the triangle is half as spread.

Example 1.10

With two tosses and a prize of X^2 (in thousands of won) for X heads, $E(X^2) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = 1.5$, so the expected prize is 1,500 won.

1.5 Distribution functions and generating functions

The **cumulative distribution function** (cdf) is $F(x) = P(X \le x)$. For discrete X it is a step function with jumps $f(x_k)$ at the x_k; for continuous X, $F(x) = \int_{-\infty}^{x} f$ and $F'(x) = f(x)$ where f is continuous.

Theorem 1.6 (Properties of a cdf)

(a) F is nondecreasing.
(b) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$.
(c) F is right-continuous.
(d) $F(a-) = P(X < a)$, hence $F(a) - F(a-) = P(X = a)$.

Proof. (a) $\{X \le x_1\} \subseteq \{X \le x_2\}$.

(b) $\{X \le -k\} \downarrow \emptyset$ and $\{X \le k\} \uparrow S$; apply Theorem 1.2.

(c) $\{X \le x + \frac{1}{n}\} \downarrow \{X \le x\}$.

(d) $\{X \le a - \frac{1}{n}\} \uparrow \{X < a\}$. ■

Conversely every function with (a)–(c) is the cdf of some random variable.

Example 1.11 (A mixed distribution)

A commuter flips a coin: on heads they wait for a shuttle that comes every 10 minutes, so the trip takes $10 + U$ minutes with $U \sim U(0, 10)$; on tails they walk, which takes exactly 15

minutes. The trip time X has

$$F(x) = \tfrac{1}{2}P(10+U \le x) + \tfrac{1}{2}I(x \ge 15) = \begin{cases} 0, & x < 10, \\ (x-10)/20, & 10 \le x < 15, \\ x/20, & 15 \le x < 20, \\ 1, & x \ge 20, \end{cases}$$

with a jump of $\frac{1}{2}$ at 15: neither discrete nor continuous.

Probability generating function. For X with values in $\{0,1,2,\dots\}$, $G(s) = E(s^X) = \sum_x s^x f(x)$ for $|s| < 1$. Since a power series determines its coefficients, $f(k) = G^{(k)}(0)/k!$ and G determines the distribution.

Moment generating function. If $E(e^{tX}) < \infty$ for all t in some $(-h,h)$, $h > 0$, the mgf is $M(t) = E(e^{tX})$; $m_k = E(X^k)$ is the kth moment.

Theorem 1.7 (Properties of the mgf)

If M exists on $(-h,h)$:
(a) all moments exist, $E(X^k) = M^{(k)}(0)$ and $M(t) = \sum_{k\ge 0} \frac{E(X^k)}{k!}t^k$ near 0;
(b) if $M_X = M_Y$ on a neighborhood of 0, then X and Y have the same distribution.

Proof of (a). Fix $|t| < h$. Since $|tx|^k/k! \le e^{|tx|} \le e^{tx} + e^{-tx}$, $|t|^k E|X|^k/k! \le M(t) + M(-t) < \infty$, so all moments exist. By Taylor's theorem with integral remainder,

$$\Big|e^{tx} - \sum_{k=0}^{n-1} \frac{(tx)^k}{k!}\Big| = \Big|\frac{(tx)^n}{(n-1)!}\int_0^1 (1-u)^{n-1} e^{utx}\,du\Big| \le \frac{|tx|^n}{n!}e^{|tx|}.$$

Since $\frac{|tx|^n}{n!} = 2^{-n}\frac{|2tx|^n}{n!} \le 2^{-n}e^{2|tx|}$, the right side is at most $2^{-n}e^{3|tx|} \le 2^{-n}(e^{3tx} + e^{-3tx})$. Taking expectations, for $|3t| < h$,

$$\Big|M(t) - \sum_{k<n} \frac{E(X^k)}{k!}t^k\Big| \le 2^{-n}\{M(3t) + M(-3t)\} \to 0.$$

So M equals its power series near 0; differentiating term by term gives $M^{(k)}(0) = E(X^k)$.

(b) is proved with characteristic functions and is beyond these notes. ■

Existence of all moments does not imply existence of the mgf (the lognormal distribution is a counterexample).

Example 1.12

For the coin-until-head variable of Example 1.7: $G(s) = \sum_k (s/2)^k = \frac{s}{2-s}$, and $M(t) = G(e^t) = \frac{e^t/2}{1-e^t/2}$ for $t < \log 2$. Then $M'(0) = 2$: on average two tosses.

Example 1.13 (Standard exponential)

For $f(x) = e^{-x}$, $x > 0$: $M(t) = \int_0^\infty e^{-(1-t)x}\,dx = (1-t)^{-1} = \sum_k t^k$ for $|t| < 1$, so $E(X^k) = k!$. (Also directly by repeated integration by parts.)

Example 1.14 (Recovering a distribution from moments)

Suppose the mgf exists and $E(X^k) = 3^k k!$. Then $M(t) = \sum_k (3t)^k = (1-3t)^{-1}$ for $|t| < \frac{1}{3}$. Since $\frac{1}{3}e^{-x/3}$ $(x > 0)$ has exactly this mgf, uniqueness shows X has this density.

Cumulants. $C(t) = \log M(t) = \sum_{r\geq 1} c_r \frac{t^r}{r!}$ defines the cumulants c_r. Expanding $\log(1+A) = A - \frac{A^2}{2} + \frac{A^3}{3} - \cdots$ with $A = m_1 t + m_2 \frac{t^2}{2!} + \cdots$ and collecting powers of t gives $c_1 = m_1 = \mu$ and $c_2 = m_2 - m_1^2 = \sigma^2$. For the standard exponential, $C(t) = -\log(1-t) = \sum_r t^r/r$, so $c_r = (r-1)!$; for Example 1.14, $c_r = (r-1)!3^r$.

1.6 Inequalities

Theorem 1.8 (Jensen)

If ϕ is convex on an interval I containing the values of X, and EX exists, then $\phi(EX) \leq E\phi(X)$.

Proof for twice differentiable ϕ. Let $\mu = EX \in I$. Taylor's theorem gives $\phi(x) = \phi(\mu) + \phi'(\mu)(x-\mu) + \frac{1}{2}\phi''(\xi)(x-\mu)^2$ with ξ between x and μ; since $\phi'' \geq 0$, $\phi(x) \geq \phi(\mu) + \phi'(\mu)(x-\mu)$ for all $x \in I$. Take expectations. (For general convex ϕ use a supporting line at μ.) ■

E.g. $(EX)^2 \leq E(X^2)$ and $E\log X \leq \log EX$ for $X > 0$.

Theorem 1.9 (Lyapunov)

If $E|X|^s < \infty$ and $0 < r < s$, then $(E|X|^r)^{1/r} \leq (E|X|^s)^{1/s}$.

Proof. For $p = s/r > 1$, $\phi(y) = y^p I(y \geq 0)$ is convex. $Y = |X|^r$ is integrable, since $|X|^r \leq I(|X| \leq 1) + |X|^s I(|X| > 1) \leq 1 + |X|^s$. Jensen gives $(EY)^p \leq E(Y^p) = E|X|^s$. ■

In particular, if the sth moment exists, all lower moments exist.

Theorem 1.10 (Markov and Chebyshev)

(a) If $E|Z|^r < \infty$ $(r > 0)$, then $P(|Z| \geq k) \leq E|Z|^r/k^r$ for every $k > 0$.
(b) If $\operatorname{Var} X < \infty$, then $P(|X - EX| \geq k) \leq \operatorname{Var} X/k^2$.

Proof. (a) $P(|Z| \geq k) = E\, I(|Z| \geq k)$, and pointwise $I(|Z| \geq k) \leq (|Z|/k)^r I(|Z|/k \geq 1) \leq |Z|^r/k^r$.

(b) Take $Z = X - EX$ and $r = 2$. ■

Example 1.15

Whatever the distribution, $P(|X-\mu| \geq 6\sigma) \leq 1/36 \approx 0.028$; for a normal population the exact value is about 2×10^{-9}. Chebyshev is universal but crude.

Theorem 1.11 (Zero variance)

If $\operatorname{Var} X = 0$ then $P(X = \mu) = 1$.

Proof. $\{|X-\mu| > 0\} = \bigcup_n \{|X-\mu| \geq \frac{1}{n}\}$, and by Chebyshev each event has probability at most $\operatorname{Var} X \cdot n^2 = 0$. Countable subadditivity gives $P(|X-\mu| > 0) = 0$. ■